

Extension of the GeN-Foam neutronic solver to SP₃ analysis and application to the CROCUS experimental reactor



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ABSTRACT

The Laboratory for Reactor Physics and Systems Behaviour at the PSI and at the EPFL has been developing since 2013 a multi-physics platform for coupled reactor analysis named GeN-Foam. The developed tool includes a solver for the eigenvalue and transient solution of multi-group neutron diffusion equations. Although frequently used in reactor analysis, the diffusion theory shows some limitations for core configurations involving strong anisotropies, which is the case for the CROCUS research reactor at the EPFL. The use of an SP₃ approximation to neutron transport can often lead to visible improvements in a code predictive capabilities, especially for one-directional anisotropies, with acceptable added computational cost vs diffusion. Following some modelling issues for the CROCUS reactor, and in order to improve the GeN-Foam modelling capabilities, the GeN-Foam diffusion solver has been extended to allow for SP₃ analyses. The present paper describes such extension and a preliminary verification using a mini-core PWR benchmark. The newly developed solver is then applied to the analysis of the CROCUS experimental reactor and results are compared to Monte Carlo calculations, as well as to the results of the diffusion solver.

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1. Introduction

Several efforts have been spent in recent years for the modernization of the numerical toolset available in the nuclear engineering community. This is witnessed by important initiatives like the CASL project (Hursin et al., 2008; Yan et al., 2011; CASL, 2011) in the US and the NURESAFE project (NURESAFE, 2013) in Europe. Some initiatives have also been undertaken aiming at a paradigm shift in the nuclear code development, the main idea being a code development based on modern numerical libraries and methodologies and not on legacy codes. Two main examples in this sense are the MOOSE project in the US (Gaston et al., 2009) and the several initiatives dedicated to the use of the CFD-oriented OpenFOAM[®] library (Weller et al., 1998; OpenFOAM, 2016) for nuclear reactor applications (Aufiero, 2014; Aufiero et al., 2014; Clifford and Ivanov, 2010; Clifford, 2013; Clifford et al., 2013; Jareteg et al., 2014, 2015). In this framework, the Laboratory for Reactor Physics and Systems Behaviour at the PSI and at the EPFL has started developing new tools for reactor analysis and has decided to privilege the use of OpenFOAM mainly due to its

open-source philosophy, which is expected to promote collaborative work and support educational efforts. A main result of this research work has been the development of a multi-physics tool for steady-state and transient analysis of nuclear reactors named GeN-Foam (Fiorina et al., 2015a,b,c, 2016). A main objective of this tool is to provide enough flexibility to analyze non-conventional reactor designs and specific phenomena that legacy codes cannot easily investigate. Such a flexibility derives from: the use of unstructured meshes allowing for fully arbitrary geometries; and the high-level object-oriented programming style of OpenFOAM, which allows for easy and relatively fail-safe modifications of the source code. A second important objective is to promote collaborative efforts on research and education, which derives naturally from the open-source philosophy associated to OpenFOAM, and from the possibility to easily develop and share stand-alone solvers and routines (in the form of C++ classes). It is worth mentioning that OpenFOAM is based on finite-volume discretization techniques and preferentially relies on operator-splitting for the coupling of different equations. This poses some limitations if compared to a finite-element framework with arbitrary coupling like that of MOOSE. On the other hand, finite volumes and operator-splitting are well-known and relatively easy techniques, which can promote the creation of a wider user and developed

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Nomenclature

Latin symbols

C_k	concentration of the k^{th} delayed neutron precursor group [m^{-3}]
D_i	neutron diffusion coefficient for the i^{th} energy group [m]
k_{eff}	effective multiplication factor [–]
S_d	delayed neutron source [$\text{m}^{-3} \cdot \text{s}^{-1}$]
$S_{n,i}$	fission neutron source from neutron energy groups others than the i^{th} [$\text{m}^{-3} \cdot \text{s}^{-1}$]
$S_{s,i}$	scattering neutron source from neutron energy groups others than the i^{th} [$\text{m}^{-3} \cdot \text{s}^{-1}$]
t	time [s]
$\mathbf{u}$	fuel velocity [$\text{m} \cdot \text{s}^{-1}$]
v_i	average neutron velocity for the i^{th} energy group [$\text{m} \cdot \text{s}^{-1}$]

Greek symbols

α	albedo coefficient [–]
β_k	delayed neutron fraction for the k^{th} delayed neutron precursor group [–]

β_t	total delayed neutron fraction [–]
λ_k	decay constant for the k^{th} delayed neutron precursor group [s^{-1}]
ν	average number of neutrons per fission [–]
$\Sigma_{a,i}$	absorption cross section for the i^{th} energy group [m^{-1}]
$\Sigma_{r,i}$	removal (disappearance) cross section for the i^{th} energy group [m^{-1}]
$\Sigma_{f,i}$	fission cross section for the i^{th} energy group [m^{-1}]
$\Sigma_{j \rightarrow i}$	group-transfer cross section from the j^{th} to the i^{th} energy group [m^{-1}]
$\varphi_{0,i}$	0 th moment of the neutron flux for the i^{th} energy group [$\text{m}^{-2} \cdot \text{s}^{-1}$]
$\hat{\varphi}_{0,i}$	flux defined in Eq. (1) [$\text{m}^{-2} \cdot \text{s}^{-1}$]
$\varphi_{2,i}$	2 th moment of the neutron flux for the i^{th} energy group [$\text{m}^{-2} \cdot \text{s}^{-1}$]
$\chi_{d,i}$	delayed neutron yield for the i^{th} energy group [–]
$\chi_{p,i}$	prompt neutron yield for the i^{th} energy group [–]

community. In addition, the large OpenFOAM user community continuously develops and shares several new solvers and routines for various applications and problems, which adds to the reliable, high-performing tools for thermal-hydraulics and thermal-mechanics that are already available in the standard OpenFOAM release.

The GeN-Foam solver tightly couples a fine/coarse-mesh sub-solver for single-phase thermal-hydraulics, a displacement-based sub-solver for thermal-mechanics, a simple sub-solver for fuel temperature prediction, and a diffusion-based sub-solver for neutronics. These features provide basic capabilities for the transient analysis of single-phase reactors. However, the diffusion approximation for neutron transport can be unsuited to analyze core configurations featuring strongly anisotropic fluxes. The important limitations associated to a diffusion approximation have been for instance observed in the modelling of the CROCUS research reactor at the EPFL (Rais et al., 2015). Aiming at a better modelling of the CROCUS reactor and at a general improvement of the GeN-Foam predictive capabilities, GeN-Foam has been modified to allow for the possibility to evaluate neutron fluxes based on an SP_3 methodology (Azmy and Sartori, 2010). An SP_3 solver can in fact provide significant improvements in a code predictive capabilities if compared to diffusion, with only a mild impact on computational requirements (IAEA-TECDOC-1700, 2013; Lee et al., 2015; NEA, 2016; Zhong et al., 2003).

Compared to most legacy codes (see e.g. (Rohde et al., 2016; PARCS, 2009)), the SP_3 solver developed for GeN-Foam has the specificity of employing general unstructured meshes. On one hand, this negatively impacts code performances, as it forces to use relatively general solution algorithms. On the other hand, it offers the possibility to study unconventional core geometries, as well as to better investigate specific phenomena, such as the reactivity effect of non-uniform core deformations in fast reactors (Fiorina et al., 2015a).

Main objectives of this paper are to describe in details and preliminary verify the extension of GeN-Foam to SP_3 analyses (Sections 2 and 3), as well as to test its new predictive capabilities via application to the CROCUS research reactor (Section 4). In particular, the results obtained using the GeN-Foam diffusion and SP_3 solvers are compared between each other and with

the results obtained using the Serpent Monte Carlo code (Leppänen, 2007). The conclusions of the work are drawn in Section 5.

2. Solver description

The SP_3 model has been implemented in GeN-Foam as an extension of the available diffusion model described in Refs. (Fiorina et al., 2015b, 2016). Structure and main features of the code have been maintained, but the equations and boundary conditions have been modified to allow for an optional transition from diffusion to SP_3 equations.

2.1. Mathematical model

Spherical harmonics (P_N) approximation (Azmy and Sartori, 2010) of the neutron transport equation is obtained via expansion of the angular dependence of the neutron flux in the first N spherical harmonics. In three dimensions, this results in a complex system of differential equations, with a number of equations growing as $(N+1)^2$. The simplified spherical harmonics (SP_N) approximation has been introduced as a trade-off between accuracy of results and complexity of the model. It was first derived heuristically (Gelbard, 1962) by replacing second derivatives with Laplacian operators in a 1-D planar formulation of the P_N approximation. In a 1-D planar geometry, and thus in the SP_N approximation, the system of equations to be solved is limited to $(N+1)/2$ elliptic (diffusion-like) equations, which provide a significant simplification over a standard P_3 approximation. In the specific case of the SP_3 approximation, 4 equations can be obtained for the 4 moments φ_0 to φ_3 , and substitution of odd-order moments into even-moment equations then allows to reduce to 2 the number of equations to be solved. It is worth mentioning that the same 2 equations have also been derived rigorously by Brantley and Larsen (2000) on the basis of variational methods using a trial function featuring a one-dimensional variation in an arbitrary direction, which provides a sounder theoretical foundation to the SP_3 methodology. Based on both the heuristic and the more rigorous derivations, it is evident that an SP_3 model can generally be considered as accurate for systems in which flux variations near

interfaces are mainly one dimensional in character, which is often the case in nuclear reactors and which explains the relatively wide success of the SP₃ methodology in the nuclear reactor community (IAEA-TECDOC-1700, 2013; Lee et al., 2015; NEA, 2016; Zhong et al., 2003).

The two equations that are derived following the two mentioned methodologies are rarely employed in their original form. One additional approximation is typically included by considering as isotropic the scattering between groups, thus obtaining the so-called within-group form of the SP₃ equations. In addition, a new variable is defined as:

$$\hat{\varphi}_0 = \varphi_0 + 2\varphi_2 \quad (1)$$

that finally allows to obtain, for each energy group i , two equations in the form:

$$\begin{aligned} \frac{1}{v_i} \frac{\partial \hat{\varphi}_{0,i}}{\partial t} = & \nabla \cdot D_i \nabla \hat{\varphi}_{0,i} + \frac{v \Sigma_{f,i} (1 - \beta_t) \chi_{p,i}}{k_{eff}} \hat{\varphi}_{0,i} - \Sigma_{r,i} \hat{\varphi}_{0,i} \\ & + \frac{S_{n,i} (1 - \beta_t) \chi_{p,i}}{k_{eff}} + S_d \chi_{d,i} + S_{s,i} + 2 \Sigma_{r,i} \varphi_{2,i} + 2 \frac{1}{v_i} \frac{\partial \varphi_{2,i}}{\partial t} \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{3}{v_i} \frac{\partial \varphi_{2,i}}{\partial t} = & \frac{3}{7} \nabla \cdot \frac{1}{\Sigma_{t,i}} \nabla \varphi_{2,i} - \left(\frac{5}{3} \Sigma_t + \frac{4}{3} \Sigma_{r,i} \right) \varphi_{2,i} + \frac{2}{3} \Sigma_{r,i} \hat{\varphi}_{0,i} - \frac{2}{3} \\ & \times \frac{v \Sigma_{f,i} (1 - \beta_t) \chi_{p,i}}{k_{eff}} \hat{\varphi}_{0,i} - \frac{2}{3} \frac{S_{n,i} (1 - \beta_t) \chi_{p,i}}{k_{eff}} - \frac{2}{3} S_d \chi_{d,i} \\ & - \frac{2}{3} S_{s,i} + \frac{2}{3} \frac{1}{v_i} \frac{\partial \hat{\varphi}_{0,i}}{\partial t} \end{aligned} \quad (3)$$

where the source terms $S_{n,i}$, $S_{d,i}$ and $S_{s,i}$ are calculated as:

$$S_{n,i} = \sum_{j \neq i} v \Sigma_{f,j} \varphi_{0,j} \quad (4)$$

$$S_d = \sum_k \lambda_k C_k \quad (5)$$

$$S_{s,i} = \sum_{j \neq i} \Sigma_{j \rightarrow i} \varphi_{0,j} \quad (6)$$

Different values can be assigned to prompt and delayed neutrons yields. The user can chose whether to use effective delayed neutron fractions and total neutron yield, or physical fractions and differentiated prompt/delayed neutron yields. It is worth noting that Eqs. (2)–(6) revert back to a standard diffusion model when $\varphi_{2,i}$ is set to zero, a characteristic that has been employed to make the use of an SP₃ model in GeN-Foam a user-selectable option. The precursor concentrations C_k in Eq. (3) are evaluated using the following equations:

$$\frac{\partial C_k}{\partial t} + \nabla \cdot (\mathbf{u} C_k) = \frac{\beta_{eff,k} \sum_j v \Sigma_{f,j} \varphi_{0,j}}{k_{eff}} - \lambda_k C_k \quad (7)$$

Eq. (5) includes a precursor transport term based on a velocity field $\mathbf{u}$, which is essential for neutronics calculations in case of liquid fueled reactors like the molten salt reactors (Fiorina et al., 2014).

2.2. Boundary conditions

The zero gradient and the fixed value boundary conditions available in OpenFOAM (OpenFOAM, 2016) can conveniently be used in both the neutron diffusion and the SP₃ solvers. An additional boundary condition has also been implemented to simulate albedo boundary conditions, i.e., boundary conditions based on the ratio α_i between incoming and outgoing partial neutron currents.

By employing the SP₃ expression of partial currents, one can demonstrate this condition to be:

$$D_i \nabla \hat{\varphi}_{0,i} = -\frac{1}{2} \left(\frac{1 - \alpha_i}{1 + \alpha_i} \right) \left(\hat{\varphi}_{0,i} - \frac{3}{4} \varphi_{2,i} \right) \quad (8)$$

$$\frac{3}{7} \frac{1}{\Sigma_{t,i}} \nabla \varphi_{2,i} = \frac{1}{2} \left(\frac{1 - \alpha_i}{1 + \alpha_i} \right) \left(\frac{3}{20} \hat{\varphi}_{0,i} - \frac{21}{20} \varphi_{2,i} \right) \quad (9)$$

2.3. Implementation details

As mentioned, the SP₃ solver for GeN-Foam has been derived as an extension of the available diffusion solver described in details in (Fiorina et al., 2016). As such, it inherits its main features in terms of implementation:

- The discretization and the solution of the equations are achieved based on the standard finite volume discretization methodology provided by OpenFOAM, which operates on unstructured meshes and optionally includes correctors for non-orthogonal meshes. In particular, a Gaussian discretization is generally used in the neutronic solver for the laplacian operator, with harmonic interpolation for the diffusion coefficients.
- It is designed both for time dependent and eigenvalue calculations. In the case of eigenvalue calculations, a traditional power iteration algorithm (Stacey, 2007) is used. In the case of transient calculations, the equations for the different energy groups are solved sequentially and the explicit terms are resolved at each time step via Picard iteration.
- Time dependent solutions can be accelerated based on three different techniques, namely: the traditional Aitken acceleration; a flux prediction based on an integral neutron balance; an “implicit predictor” that couples all energy groups in each cell by fixing the leakage term.
- It is provided with the possibility to use isotropic discontinuity factors, and with a procedure to automatically adjust the discontinuity factors based on fluxes from full-core Serpent calculations. The same discontinuity factor is used for both moments in the SP₃ equations.
- Different cross sections are provided to different core regions based on a fully arbitrary mesh subdivision in cell zones (i.e., groups of labelled cells). This allows a definition of the cross sections at a region, assembly or pin level.
- Cross section parametrization is performed for each cell based on local fuel temperature, fuel density, coolant temperature, coolant density, fuel axial expansion, core radial expansion, cladding expansion, boron concentration. A linear interpolation is performed in all cases except for fuel temperatures, for which the interpolation can be based either on the logarithm or on the square root of temperature, based on whether the neutron spectrum is mainly fast or thermal, respectively.
- It is provided with the possibility to deform the mesh based on a given displacement field to account for thermal deformation of a core, which allows for an accurate prediction of expansion-related reactivity feedbacks (Fiorina et al., 2015a).

3. Solver verification

A code-to-code comparison has been performed against PARCS (PARCS, 2009) to verify the accuracy and performances of the developed SP₃ model. It is worth noting that PARCS solves the same analytical model as GeN-Foam. In the expected case of consistent discretization schemes, identical results are thus expected for the two codes.

As a test case, a PWR mini-core (Sjenitzer et al., 2015) derived from the OECD/NEA benchmark presented by (Kozłowski and Downar, 2003) has been considered. It consists of a group of 9 fuel assemblies: a central UO₂ assembly surrounded by 8 MOX assemblies. A water reflector with a thickness equal to that of the assemblies is placed radially around the 9 fuel assemblies. A control rod is inserted in the upper part of the UO₂ central assembly. Fig. 4 shows a cut of the geometry, while details about dimensions and materials can be found in (Sjenitzer et al., 2015). The two-groups cross sections have been generated based on full core Serpent calculations, with universes defined as in Fig. 1, namely: water reflector; corner MOX assemblies in the rodged part; side MOX assemblies in the rodged part; central UO₂ assembly in the rodged part; corner MOX assemblies in the unrodged part; side MOX assemblies in the unrodged part; and central UO₂ assembly in the unrodged part. Multiplication factor predicted by Serpent is equal to $0.99992 \pm 3.8 \cdot 10^{-5}$.

The SP₃ model implemented in PARCS assumes that $\Sigma_t = \Sigma_{tr}$ in Eq. (3). This is an approximation that allows to use the same cross-section set as for the diffusion model, where Σ_t is normally not used. For a proper comparison, a modified version of the GeN-Foam model has thus been prepared assuming $\Sigma_t = \Sigma_{tr}$. At first, the correct implementation of the model has been verified based on a simplified 1-D geometry made of two infinite planes of thicknesses equal to the assembly pitch (21.42 cm), and using the cross-sections for the unrodged UO₂ assembly and for the water reflector. In the case of reflective boundary conditions, the same k_{eff} have been predicted by the two codes for both the SP₃ and the diffusion models (1.02015 and 1.01531 for the SP₃ and the diffusion models, respectively). If void boundary conditions (corresponding to albedo boundary conditions with zero incoming current) are employed on the reflector side, both codes predict the same k_{eff} (equal to 1.01479) when using a diffusion approximation, while GeN-Foam predicts a 4 pcm lower k_{eff} (1.01933 instead of 1.01937) when using an SP₃ approximation, which suggests some differences in the implementation of boundary conditions. Deeper investigation of the observed discrepancy has not led to the identification of possible implementation mistakes in GeN-Foam, but the issue will be subject of further investigation in the future.

The full mini-core PWR benchmark was then simulated to evaluate the different performances of the two codes. A relatively fine mesh was necessary in both codes (~ 3 mm per cell for GeN-Foam and 4 nodes per assembly for PARCS, respectively). In case of

GeN-Foam, this led to a computationally intensive calculation requiring few tens of GB of RAM. The possibility to parallelize the computations allowed for reasonable solution times of several minutes, but it should be noted that PARCS could solve the problem in a few seconds. Intensive use of computational resources is the downside of using a numerical library based on unstructured meshes and finite-volume discretization techniques. Unstructured grids need a large memory for storing nodes, faces, volumes, and information on their connectivity. In addition, in structured meshes, data for cells that are geometrically close is also close in memory, which results in a better usage of the memory bandwidth. The size and the optimal use of memory are a critical concern for codes devoted to PDE solutions as the memory bandwidth is normally the main bottleneck (not computation power). An additional significant advantage of structured meshes is the possibility to make use of the a priori knowledge of a matrix structure. PARCS for instance makes a wise use of the recursive three level block tridiagonal structure of its matrix to rapidly build an optimal preconditioner for the solution of the linear system. Finally, when employing a structured mesh, it is often possible to employ analytical solutions at a node level to implement, as in PARCS, efficient nodal methods requiring coarser meshes. Advantages of an unstructured mesh come from the complete freedom in meshing unconventional geometries, and from the availability of general projection algorithms for transferring data between the different meshes used to solve different physical phenomena. Besides, it should be noted that in a multi-physics code like GeN-Foam the bottleneck in terms of computational requirements is typically the thermal-hydraulics, so that a relatively slow neutronics solver does not dramatically impact the overall solver performances.

It is worth mentioning that OpenFOAM allows for the use of second order discretization schemes, which is the case of the Gaussian discretization scheme with harmonic interpolation for the diffusion coefficients that is normally employed to discretize the laplacian operator in GeN-Foam. The second order spatial convergence of GeN-Foam has been successfully verified employing a simple 9-assembly 2-D geometry, with a number of cell per assemblies ranging from 64 to 65536. Results are shown in Fig. 2.

As regards the computational cost of an SP₃ vs a diffusion simulation, the number of equations to be solved in the SP₃ solver doubles compared to the diffusion solver, which should essentially double the computational time for a given mesh. As a matter of fact, the solution of the second moment equations has been observed to be typically faster compared to the zero moment equations (and thus to diffusion). As a result, in the case here considered, the SP₃ solver required 50% more computational time than diffusion. On the other hand, an SP₃ solver can sometimes require significantly finer meshes to reproduce the steep gradients of the second moment flux at interfaces.

For completeness, the GeN-Foam model has also been run for the full mini-core geometry without assuming $\Sigma_t = \Sigma_{tr}$. It was found that this assumption significantly increases the SP₃ correction in the case here considered. While using Σ_t in Eq. (3) leads to a $k_{eff} \sim 150$ pcm higher than in a diffusion approximation, the use of Σ_{tr} determines a ~ 450 pcm higher k_{eff} . In this case, the approximation related to the use of the transport cross section is thus higher than the SP₃ correction itself.

4. Application to the CROCUS experimental reactor

CROCUS is an experimental zero-power (100 W) water-moderated reactor mainly dedicated to teaching radiation and reactor physics (Paratte et al., 2006; Kasemeyer et al., 2007). The core is located in an aluminum vessel of 130 cm diameter and

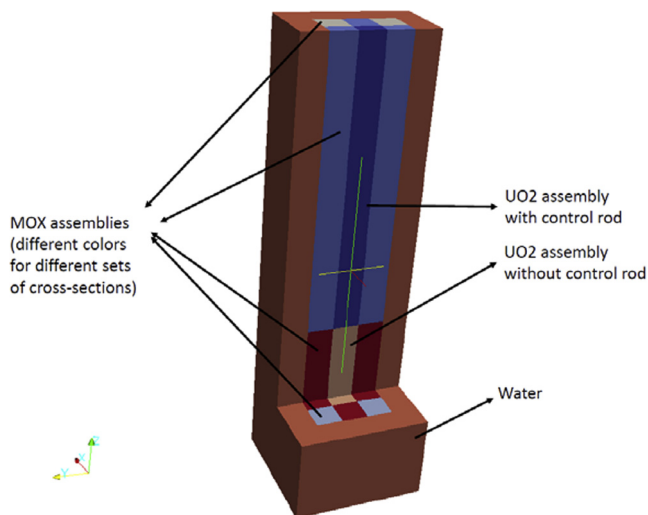


Fig. 1. Geometry of the mini-core PWR benchmark.

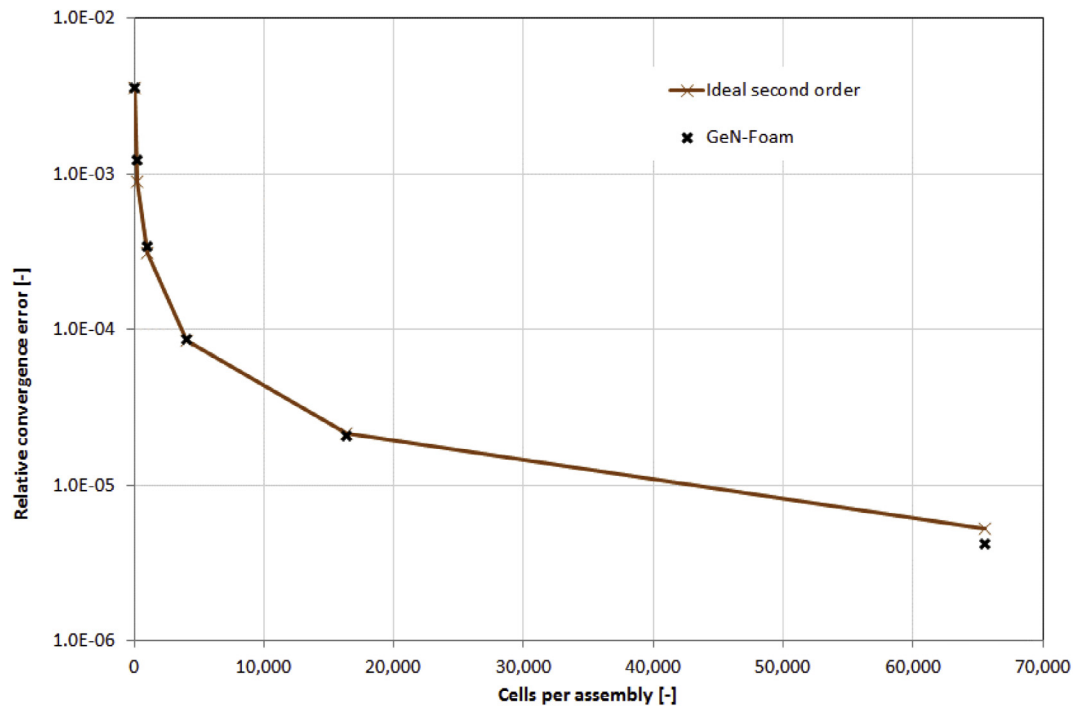


Fig. 2. Mesh convergence of the k_{eff} predicted by GeN-Foam and comparison with an ideal second order curve.

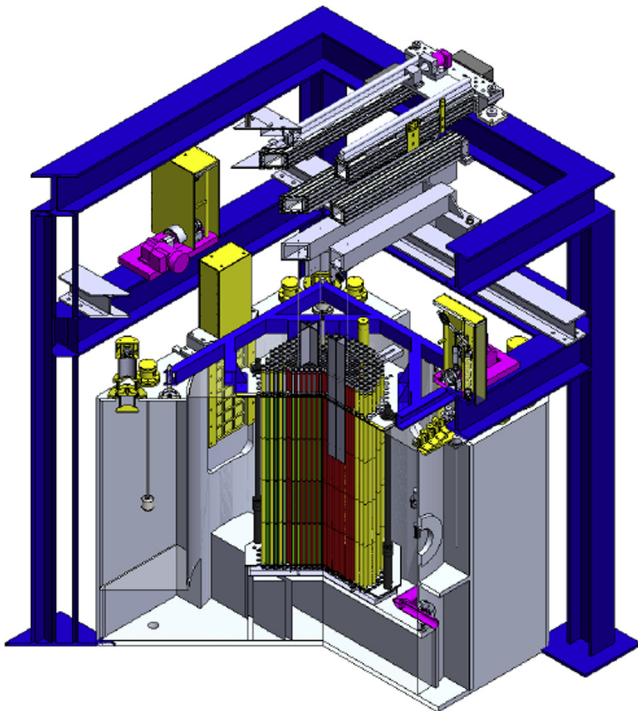


Fig. 3. Cut view of the CROCUS core.

1.2 cm thickness (see Fig. 3). It is filled with demineralised light water used as both moderator and reflector. It operates at room temperature using a controlled water loop with secondary and tertiary circuits, two heat exchangers and an electrical heater.

The core active part has the approximate shape of a cylinder of about 60 cm in diameter and 1 m in height. As shown

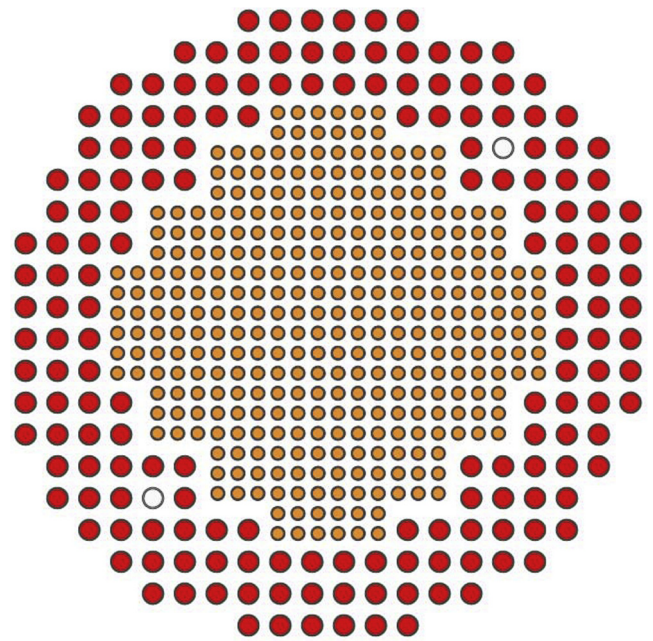


Fig. 4. Schematic top view of the CROCUS loading pattern.

schematically in Figs. 4 and 5, it consists of two fuel zones with square lattices of different pitches:

- an inner zone of 336 UO_2 rods with an enrichment of 1.806 wt.% and a pitch of 1.837 cm;
- an outer zone of 172 U_{metal} rods in nominal configuration, 0.947 wt.% and 2.917 cm.

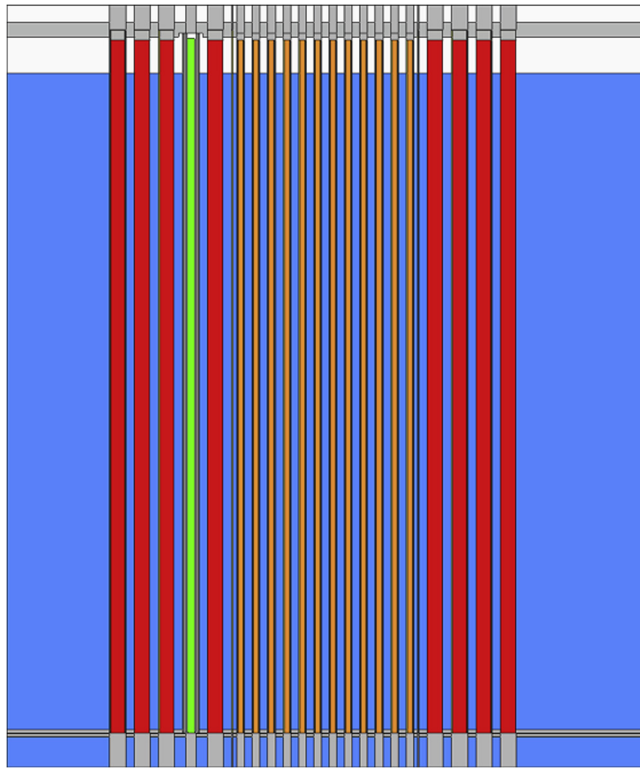


Fig. 5. Schematic lateral view of the CROCUS loading pattern (including one inserted control rod, in green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

A water gap exists between the two zones because of the two different pitches. Both uranium fuels consist of a 1-m pile of cylindrical pellets clad in aluminum.

The rods are kept in position by two octagonal aluminum grid plates spaced 1 m apart. The top plate is 2.05 cm thick while the

bottom plate is 1.05 cm thick. A 0.5-mm cadmium layer is placed in the mid-plane of each aluminum plate to limit neutron doses in the reactor cavity. The active zone of the fuel starts above the lower cadmium layer. Reactivity is controlled either by water level using a spillway, or by two B_4C absorber rods. With the current fuel loading, the critical water level is 95.22 ± 0.01 cm with both rods fully extracted. Therefore a small axial section of the active core is exposed to air at atmospheric conditions as shown in Fig. 5.

4.1. Modelling choices

Two-groups cross sections have been generated based on full core Serpent calculations, with universes defined as in Fig. 6. In particular, inner core, outer core and reflector have been divided based on the presence of water or air. These regions also include the part of the grids interior to the two Cd layers. The Cd layers and all structures external to them are axially homogenized. Control rods have been considered as half-way inserted.

Two different GeN-Foam models have been prepared. In one case, the full geometry shown in Fig. 6 has been simulated assuming void boundary conditions everywhere. In the second case (Fig. 7), only the structures placed in-between the 2 cadmium layers have been simulated, and proper albedo boundary conditions have been applied to the top and bottom boundaries. The albedo coefficients have been evaluated using Serpent and are reported in Table 1. To obtain mesh convergence, a radial discretization was used with 4 cells per pin. In the axial direction, an irregular mesh with gradual refinement towards the cadmium layers was used to correctly reproduce the steep gradient at boundaries without excessively affecting computational requirements. In both cases, the result has been a mesh of approximately 1.5 million cells and an axial size of the order of 1 mm in the proximity of the cadmium layers.

4.2. Results

Figs. 8–11 show the radial and axial flux profiles predicted by Serpent and by the GeN-Foam SP₃ solver, and include the

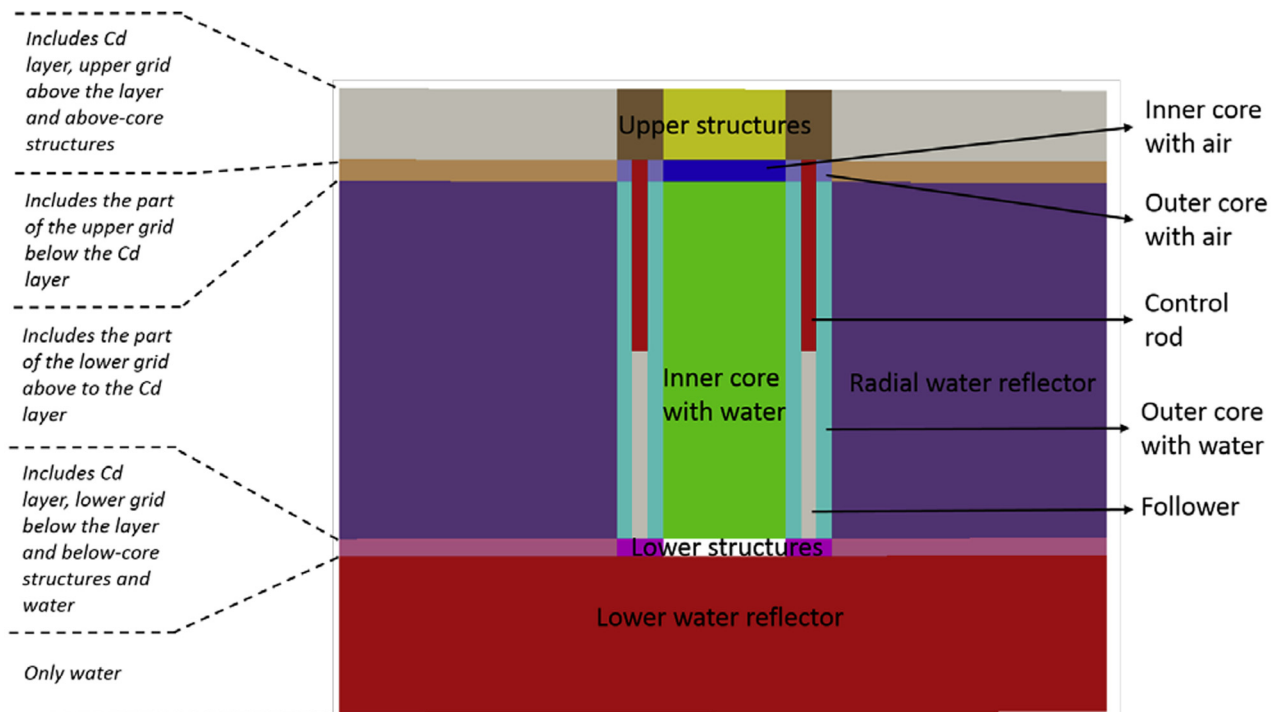


Fig. 6. Universes used in Serpent for cross-section preparation.

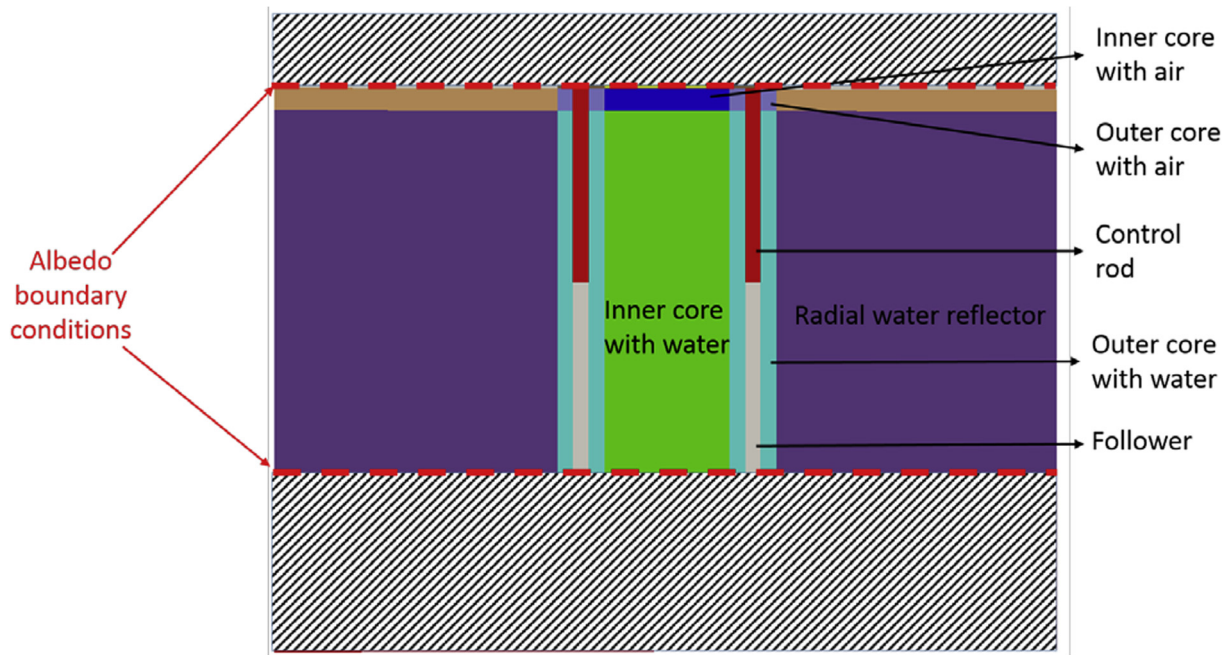


Fig. 7. Reduced geometry also used for GeN-Foam calculations.

Table 1

Albedo coefficient (α in Eqs. (8) and (9)) for top and bottom boundaries in Fig. 6.

	Fast group	Thermal group
Top boundary	0.289	0.022
Bottom boundary	0.372	0.335

difference between the SP₃ and the diffusion approximation. Differences between diffusion and SP₃ solvers are relatively small in all cases, with non-negligible differences mainly close to material interfaces and external boundaries, as expected.

Use of a full (Fig. 6) or a reduced (Fig. 7) geometry negligibly affects the radial flux profiles. On the other hand, the axial flux profiles predicted by GeN-Foam appear to be visibly improved when using a reduced geometry, as the use of a full geometry causes the flux to be slightly shifted towards the bottom of the core. This has to be ascribed to the diffusive nature of the equations, which are incapable of accurately describing the flux behaviour close to material interfaces. This is particularly the case for the lower boundary of the active core, which separates a highly multiplying core region and a highly absorbing region containing cadmium.

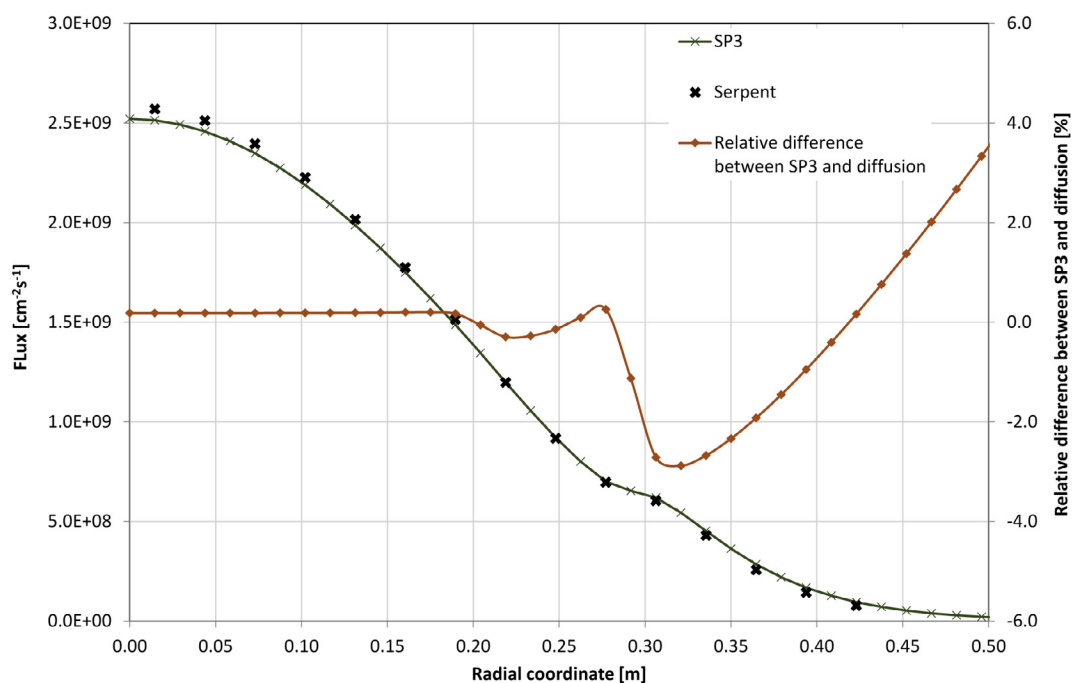


Fig. 8. Radial profiles (along x axis) for the full geometry (Fig. 6).

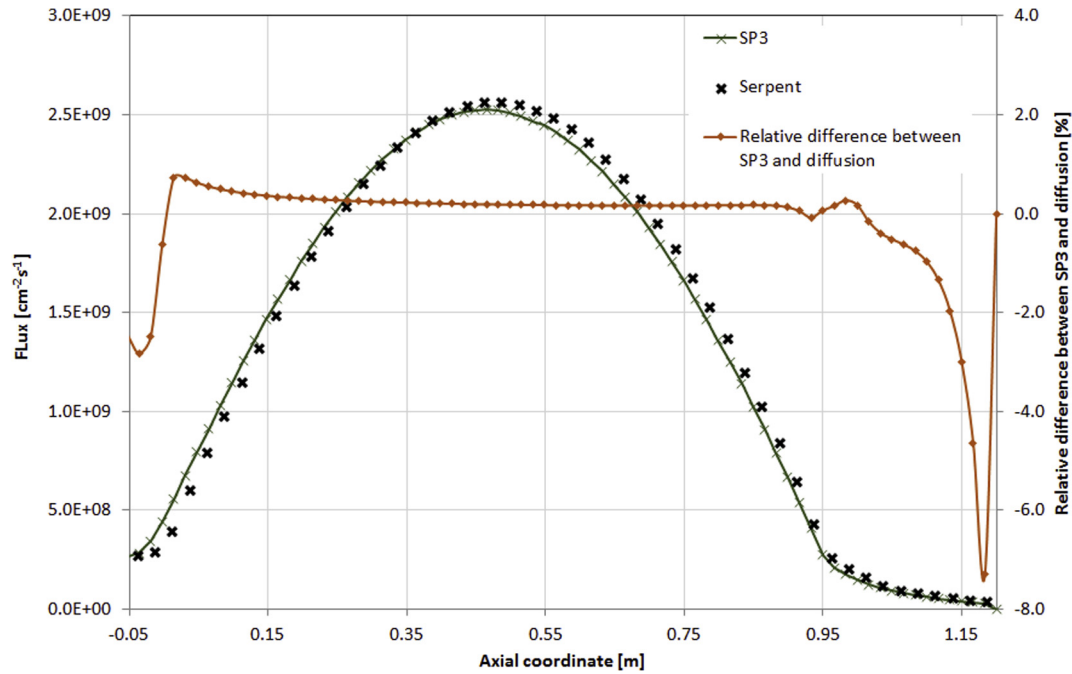


Fig. 9. Axial profiles at centerline for the full geometry (Fig. 6).

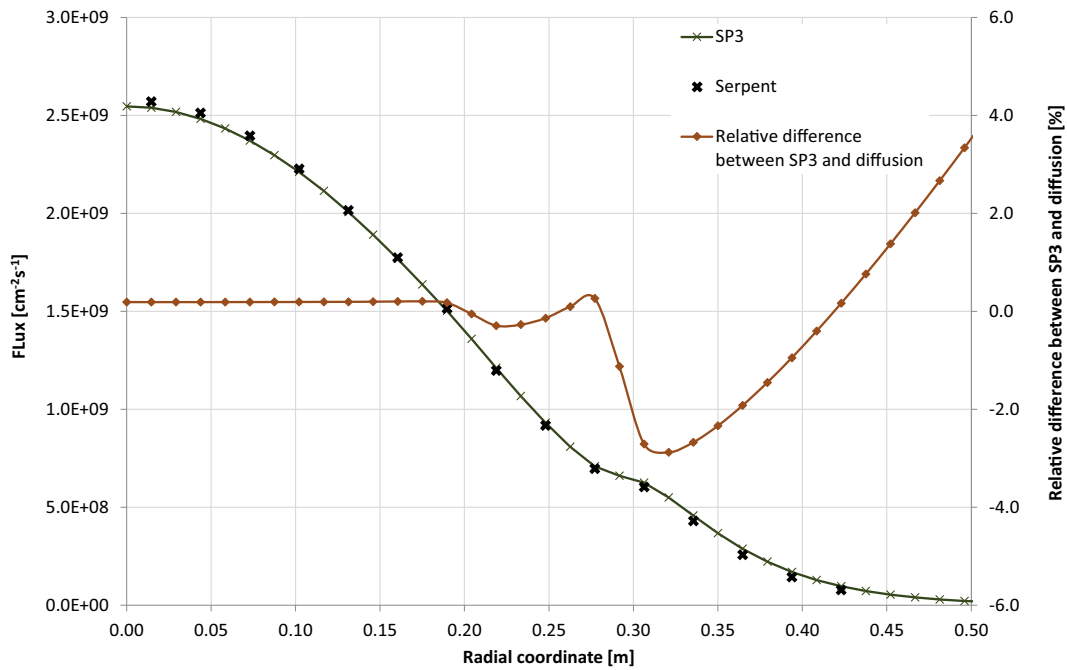


Fig. 10. Radial profiles (along x axis) for the reduced geometry (Fig. 7).

Table 2 shows the results obtained in terms of k_{eff} , control rod worth and water level worth (from the nominal 95.22 cm level to 75.22 cm). The diffusion approximation leads to an underprediction of the reactor k_{eff} by approximately 1200 pcm, which has been ascribed to the underestimation of flux in the radial water reflector. An improved (higher) prediction in terms of k_{eff} is obtained using an SP_3 approximation. This is consistent with results in Figs. 8 and 10 that show a higher predicted flux by the SP_3 solver in the core region.

The control rod worth is well reproduced by both the diffusion and the SP_3 calculations, which is related to the accurate radial flux

prediction in the outer core region, and to the relatively small flux reduction caused by the control rods (approximately 10%). However, a visibly better prediction is provided by the SP_3 model. No significant differences are observed between full and reduced geometries, which is consistent with the similar radial flux predictions.

The use of an SP_3 model tends to improve predictive capabilities also in terms of the reactivity variations associated to a variation of the water level. The full geometry shows in this case better predictions, but this has to be ascribed to a compensation of errors, since the full geometry provides visibly worse predictions in terms of flux profiles.

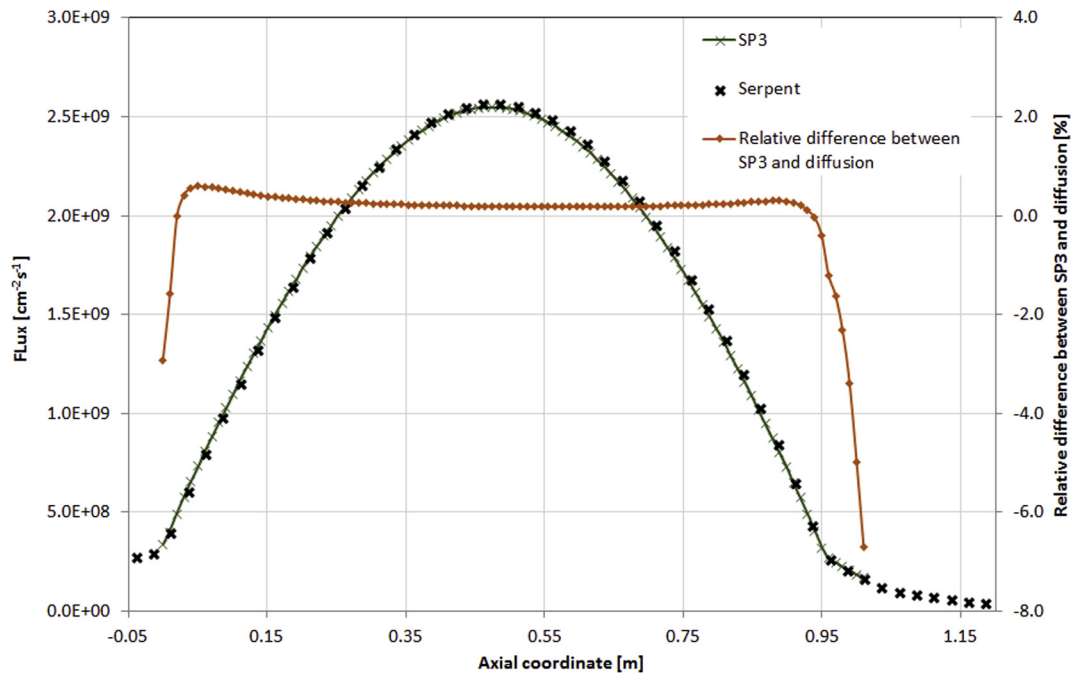


Fig. 11. Axial profiles at centerline for the reduced geometry (Fig. 7).

Table 2

Comparison between Serpent and GeN-Foam results in terms of k_{eff} , control rod worth and water level worth.

		k_{eff}		Control rod worth		Water level worth (20 cm)	
		Value [–]	dk/k w.r.t Serpent [pcm]	Value [pcm]	Difference w.r.t Serpent [%]	Value [pcm]	Difference w.r.t Serpent [%]
Serpent		1.00100 ± 1 pcm	–	347.6 ± 2 pcm	–	-1420 ± 2 pcm	–
Full geometry (Fig. 6)	Diffusion	0.98921	–1178	355.7	2.3	–1449	2.0
	SP ₃	0.99114	–986	343.9	–1.1	–1434	0.9
Reduced geometry (Fig. 7)	Diffusion	0.98875	–1224	355.3	2.2	–1541	8.5
	SP ₃	0.99070	–1029	343.6	–1.2	–1526	7.4

5. Conclusions

An SP₃ model has been implemented in the GeN-Foam multi-physics solver as an extension of the available diffusion solver. The new solver has been successfully verified against the PARCS code. However, the verification process has shown notably higher computational requirements for the newly developed solver. This follows from the choice of adopting a numerical approach based on unstructured meshes and from applying it to a regular geometry that can obviously benefit from a structured mesh. On the other hand, an unstructured approach is essential for simulating the unconventional core geometries of some innovative reactors (e.g., the molten salt reactors), as well as many research reactors. In addition, it allows for generalized routines for projecting fields between different meshes in a multi-physics solver.

After verification, the solver has been applied to the analysis of the CROCUS research reactor at the EPFL, and the results compared with diffusion calculations and Monte Carlo results. Two different modelling choices have been pursued while setting up the geometry: in one case the whole reactor cavity is simulated; in the second case the geometry is reduced axially and limited to the two Cd layers that are used as axial shields. The second choice has been found to give better results as it avoids the solver to deal with an interface featuring a very sharp change in cross sections. When using the reduced geometry, flux shapes are well reproduced by both

the diffusion and the SP₃ solvers. A small inaccuracy is observed only at the radial boundary between core and water reflector, where an overestimation of flux is observed on the reflector side. Such a small inaccuracy has an important impact on reactivity. The use of an SP₃ approximation allows for a more accurate flux distribution, which translates into significantly improved predictions in terms of reactivity, control rod worth and water level worth.

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